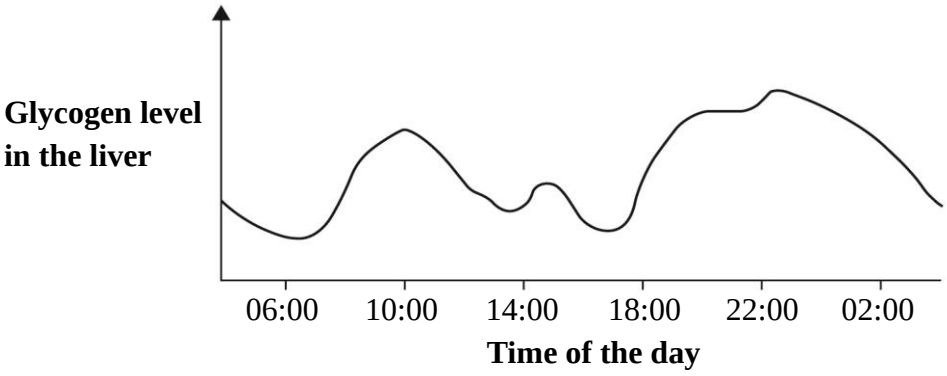
Structured questions

|!|BCELB0190000001|!| ★

Andy is a healthy male. The graph below shows the changes of the glycogen level in his liver during 24 hours.



During the 24 hours, Andy had breakfast at 06:00, lunch at 13:30, dinner at 17:30 and had a snack at 22:00.

(a) Describe and explain how the glycogen level in Andy's liver changed in the following time periods:

(i) from 06:00 to 10:00 (4 marks)

(ii) from 10:00 to 14:00 (2 marks)

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(b) Leon and Andy are of the same age and similar build, but Leon has Type 2 diabetes. If Leon and Andy ate the same foods, how would the glycogen level in Leon's liver be different from Andy's? Explain your answer. (3 marks)

##

(a) (i) After breakfast at 06:00, the glycogen level increased until 10:00. (1)
As the food eaten is digested and absorbed into the blood, the blood glucose level increased. (1)
More insulin is released from the pancreas. (1)
As a result, liver cells were stimulated to convert more glucose to glycogen which was stored in the liver. (1)

(ii) From 10:00 to 14:00, the glycogen level fell. (1)
This is because the glycogen stored in the liver was converted to glucose and released into the blood. (1)

(b) The rise in glycogen level in Leon's liver would be smaller compared with Andy's. (1)
As Leon has Type 2 diabetes, his cells are resistant to insulin. (1)
Therefore, less glucose is converted to glycogen in the liver. (1)

(9 marks)

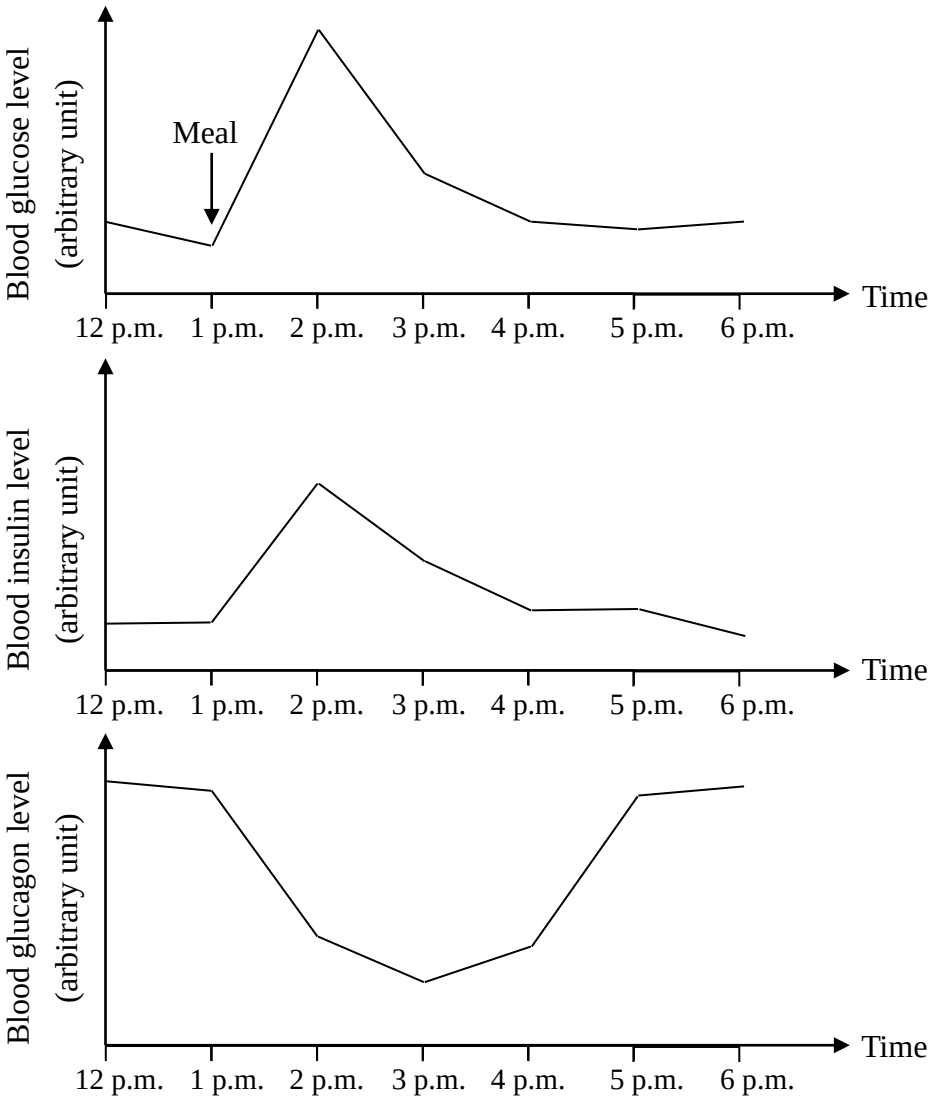
##

	Skills assessed	Marks
(a)(i)	Analysing data	1
	Understanding basic concepts	3
(a)(ii)	Analysing data	1
	Understanding basic concepts	1
(b)	Applying concepts	3

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|!|BCELB0190000002|!| ★

The following graphs show the changes in the glucose, insulin and glucagon levels in the blood of a healthy person before and after a meal rich in carbohydrates.



(a) Explain why the blood glucose level increases after 1 p.m. (2 marks)

(b) Describe and explain the changes in the blood glucose level from 2 p.m. to 3 p.m. (4 marks)

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(c) Explain why the blood glucagon level increases from 3 p.m. to 5 p.m. (2 marks)

(d) A student wrote the following paragraph.

> Insulin and glucagon are produced in the pancreas and released through the pancreatic ducts into the blood. Both are enzymes that regulate the blood glucose level. They act on the liver cells and their actions are antagonistic.

Correct the ***two*** misconceptions in the above paragraph. (2 marks)

##

(a) The food rich in carbohydrates are digested into glucose in the alimentary canal. (1)
The large amount of glucose is absorbed into the bloodstream from the small intestine. (1)

(b) The increased insulin level and the decreased glucagon level in the blood (1)
lead to the conversion of excess blood glucose to glycogen by the liver and (1)
an increased uptake of glucose by body cells for respiration. (1)
Hence, the blood glucose level decreases. (1)

(c) The blood glucose level decreases from 3 p.m. to 5 p.m. (1)
This stimulates the pancreas to release more glucagon. (1)

(d) Insulin and glucagon are released directly into the blood from the pancreas, but not through the pancreatic duct. (1)
Insulin and glucagon are hormones, but not enzymes. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Analysing data	1
	Applying concepts	3

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(c)	Applying concepts	2
(d)	Applying concepts	2

|!|BCELB0190000003|!| ★ **STSE**

Sandra suffers from diabetes mellitus. She needs regular injections of hormone X to keep her blood glucose level stable.

(a) Name hormone X. (1 mark)

(b) State ***one*** effect of hormone X. (1 mark)

(c) If Sandra could not receive regular injections of hormone X, how would the water balance in her body be affected? (4 marks)

(d) The glycemic index (GI) measures the effects of carbohydrates in foods on blood glucose levels. Ingestion of high-GI foods causes a more rapid rise in the blood glucose level.

(i) Which type of drinks, milk or sports drinks, has a lower GI value? Explain your answer. (3 marks)

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(ii) Suggest why it may be beneficial for Sandra to be aware of the GI of foods.

(2 marks)

##

(a) insulin (1)

(b) Insulin stimulates liver cells to convert glucose to glycogen.

OR

Insulin stimulates body cells to take up more glucose which is oxidized in respiration. (1)

(c) Less glucose would be taken up by body cells and her blood glucose level would become very high. (1)

The water potential of her blood would decrease. (1)

Water would move out of the cells by osmosis. (1)

This leads to an increase in blood volume and hence an increase in urine production. / The decrease in water potential of the blood stimulates the 'thirst centre' in the brain, making the person thirsty. (1)

(d) (i) milk (1)

Lactose in milk has to be broken down first before absorption, while glucose in sports drinks can be absorbed directly. (1)

Drinking milk will cause a slower rise in the blood glucose level compared with consuming sports drinks. (1)

(ii) Being aware of the GI of foods can help her choose low-GI foods which cause slow rise in her blood glucose level, (1)

so that great fluctuations in her blood glucose level can be avoided. (1)

(11 marks)

##

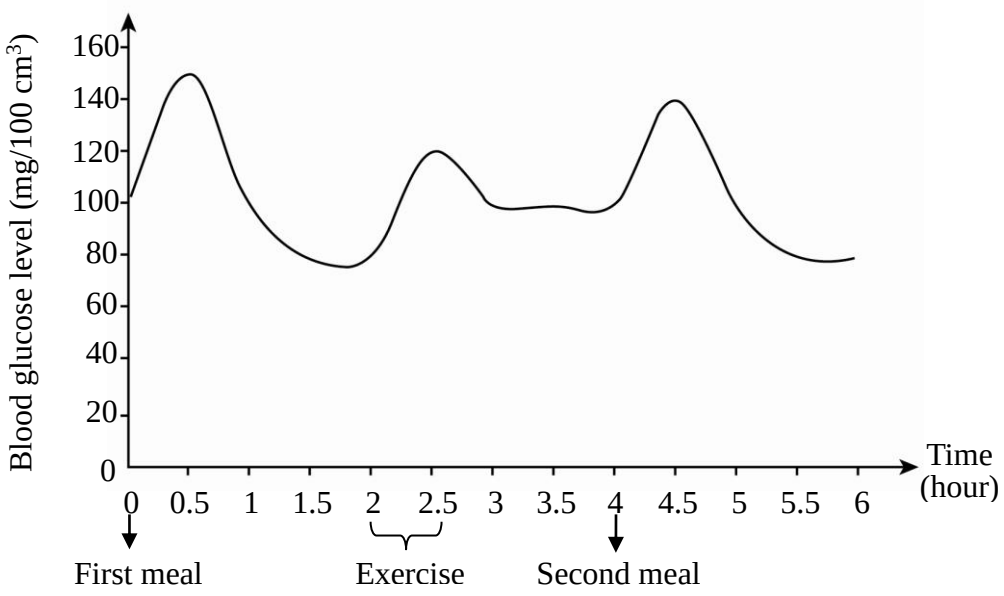
	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Applying concepts	4
(d)(i)	Applying concepts	3
(d)(ii)	Applying concepts	2

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|!|BCELB0190000004|!| ★ **SI**

Ten healthy people (with no diabetes) participated in a study which investigates how eating and exercise affect the blood glucose level and the level of insulin in blood. Graph I and Graph II below show the changes in the average values of their blood glucose level and blood insulin level.

Graph I



Graph II



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(a) Explain the rise in the blood glucose level during the 30 minutes after a meal. (2 marks)

(b) Explain the rise in the blood glucose level during exercise. (4 marks)

(c) Explain why it is better to take the average value of data from a group of participants rather than from a single participant. (2 marks)

##

(a) Carbohydrates in food are digested to glucose (1)
which is absorbed into the bloodstream in the small intestine. (1)

(b) Cells take up more glucose from the blood for respiration to release more energy for exercise. (1)
The pancreas detects the decrease in the blood glucose level (1)
and it releases more glucagon and less insulin into the blood. (1)
As a result, glucagon stimulates liver cells to convert the stored glycogen to glucose. (1)
The glucose formed is then released into the blood.

(c) Taking an average value can avoid the effect of any extreme values from a single participant. (1)
This ensures that the results are more reliable. (1)

(8 marks)

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	4
(c)	Practical skills	2

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|!|BCELB0190000005|!| ★

Two men, A and B, ate 100 g of steamed white rice. Their blood glucose levels were then monitored for three hours. The table below shows the results.

Time after eating (minutes)	**Blood glucose level (mg/100 cm^3)**	
	A	**B**
0	80	160
30	105	250
60	155	300
90	125	290
120	78	275
150	75	240
180	80	210

(a) On the graph paper provided, plot a graph to show the changes in the blood glucose levels of A and B with time. (5 marks)

(b) Explain the changes in the blood glucose level within the first hour after the ingestion of rice. (2 marks)

(c) Three hours after the ingestion of rice, urine samples were collected from A and B and were tested with Benedict's test. What results would be observed? Explain your answers. (4 marks)

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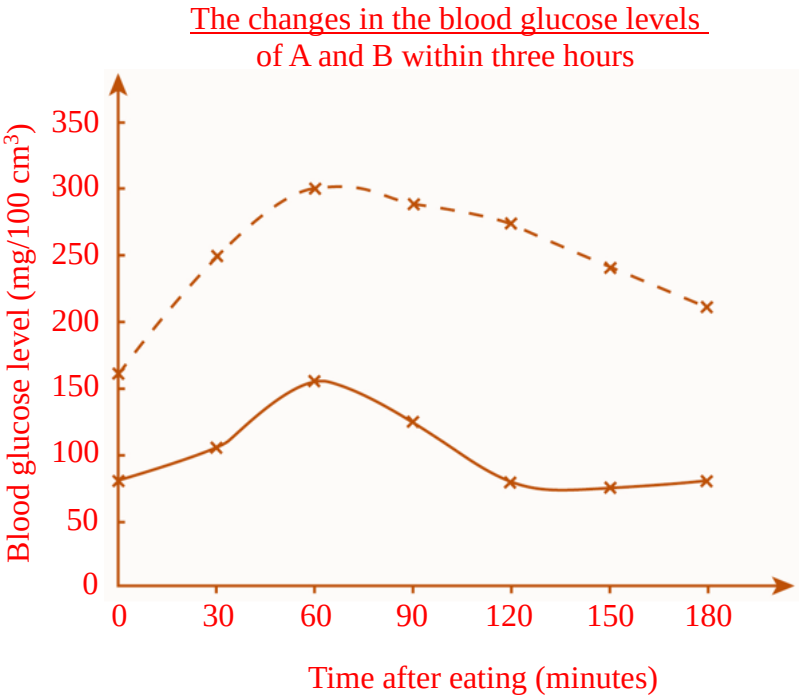
(d) Based on the information given, identify the disease that B may suffer from. Suggest a possible cause of this disease. (2 marks)

_____

_____

##

(a)



Correct title (1)
Correct choice of axes (1)
Axes with labels and units (1)
Correct plotting and joining of line (1, 1)

(b) Starch in rice is digested into glucose which is absorbed into the bloodstream at the small intestine. (1)
This leads to a rise in the blood glucose level. (1)

(c) The urine sample from A would result in a blue mixture, indicating the absence of any reducing sugars in the sample. (1)
The blood glucose level of A returned to normal after three hours. Glucose is not excreted in urine. (1)
The urine sample from B would result in red precipitates, indicating the presence of reducing sugars in the sample. (1)
The blood glucose level of B remained high after three hours. Glucose is excreted in urine. (1)

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(d) Diabetes mellitus (1)

Any ***one*** of the following causes: (1)

- The pancreas cannot secrete enough insulin.
- The body cells do not respond normally to insulin.

(Or other reasonable answers)

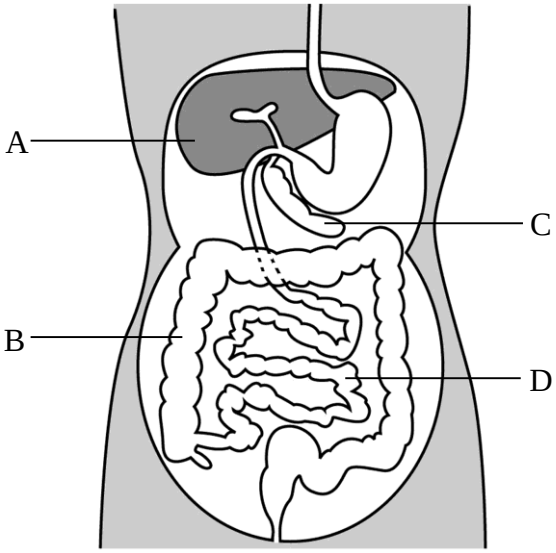
(13 marks)

##

	Skills assessed	Marks
(a)	Presenting information	5
(b)	Applying concepts	2
(c)	Applying concepts	4
(d)	Analysing data	1
	Understanding basic concepts	1

|!|BCELB0190000006|!| ★

The diagram below shows part of the human digestive system.



(a) (i) Which labelled structures are involved in the digestion of carbohydrates?

(2 marks)

(ii) Describe how the structures you have given in (i) help with the digestion of carbohydrates. (4 marks)

(b) State the roles of structure C in the regulation of blood glucose level. (3 marks)

(c) Describe how structure A can help maintain a stable blood glucose level during fasting. (2 marks)

##

(a) (i) Structures C and D (1, 1)

(ii) Structure C secretes pancreatic juice which contains amylase. (1)
Amylase catalyses the breakdown of starch to maltose (1)
The enzymes present on the cell membranes of the epithelial cells of the wall of structure D (1)
catalyse the digestion of disaccharides to monosaccharides such as glucose. (1)

(b) Structure C detects changes in the blood glucose level. (1)
Structure C secretes insulin to lower the blood glucose level and (1)
glucagon to raise the blood glucose level. (1)

(c) Under the action of glucagon, more glycogen is converted to glucose in structure A (liver). (1)
This raises the blood glucose level back to normal level. (1)

(11 marks)

##

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	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Applying concepts	4
(b)	Understanding basic concepts	3
(c)	Applying concepts	2

|!|BCELB0190000007|!| ★

Hypoglycaemia is a condition in which the blood glucose level of a person drops below the normal level.

(a) Give ***two*** symptoms caused by hypoglycaemia. (2 marks)

(b) If too much hormone X is present in the blood, hypoglycaemia may occur.

(i) What is hormone X? (1 mark)

(ii) What is the effect of hormone X? (1 mark)

(c) State and explain ***two*** possible causes of hypoglycaemia. (4 marks)

##

(a) Any ***two*** of the following: (2)

- a feeling of weakness
- fatigue
- blurred vision
- loss of consciousness

(Or other reasonable answers)

(b) (i) Insulin (1)

(ii) Hormone X stimulates body cells to take up more glucose for respiration.

OR

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Hormone X stimulates liver cells to convert more glucose to glycogen for storage. (1)

(c) Any ***two*** sets of the following: (4)

- Overdose of insulin (1)
  Insulin stimulates increased conversion of glucose to glycogen by liver cells / increased uptake of glucose by body cells. (1)
- Skipping a meal (1)
  The blood glucose level may fall when body cells continuously consume glucose in respiration but without the intake of glucose from food. (1)
- Exercising too vigorously (1)
  The blood glucose level may fall when a large amount of glucose is consumed in respiration to release energy for muscle contractions. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)(i)	Applying concepts	1
(b)(ii)	Understanding basic concepts	1
(c)	Applying concepts	4

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|!|BCELA0190000008|!| ★★ **NOS** **Cross topic**

The table below lists some key discoveries towards the identification of insulin as a treatment for diabetes.

Year	Scientists	Discovery
1869	Paul Langerhans	Under microscope examination, he observed two systems of cells in the pancreas: clusters of lightly stained cells surrounded by more darkly stained cells.
1889	Oscar Minkowski and Joseph von Mering	• After the pancreas of a dog was removed, the dog developed diabetes. • After the pancreatic duct of a dog was surgically tied off, the dog developed digestive problems but not diabetes.
1909	Eugene Opie	The clusters of lightly stained cells in the pancreas of diabetic patients had degenerated.
1921–1922	Frederick Banting and Charles Best	Banting and Best tied off the pancreatic duct of a dog. This caused the darkly stained cells in its pancreas to degenerate, leaving the lightly stained cells functional. They then removed the pancreas from the dog and ground up the tissue to obtain an extract. After the extract was injected into diabetic dogs, the symptoms of diabetes were reduced.

(a) Based on your knowledge of the functions of the pancreas, explain the findings of Minkowski and Mering. (4 marks)

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(b) Before Banting and Best, other scientists had tried to use extract from a ground-up pancreas to treat diabetes, but they were unsuccessful. Suggest an explanation for this. (2 marks)

(c) Explain why the blood glucose levels of the diabetic dogs decreased after Banting and Best injected them with the pancreas extract. (3 marks)

(d) Give ***one*** aspect about the nature of science demonstrated by the events listed in the table. (1 mark)

##

(a) The pancreas secretes insulin (and glucagon) directly into the blood to regulate the blood glucose level. (1)
The pancreas also produces pancreatic juice which passes through the pancreatic duct for food digestion. (1)
When the pancreas was removed, there is no secretion of insulin (and glucagon) to regulate the blood glucose level, so diabetes developed. (1)
When only the pancreatic duct was tied off, the secretion of insulin (and glucagon) was unaffected, but pancreatic juice could not pass into the intestine and so digestive problem developed. (1)

(b) Some cells of the pancreas produce pancreatic juice (or digestive enzymes). (1)
When the pancreas was ground, pancreatic juice (or digestive enzymes) digests the insulin. (1)

(c) The pancreas extract contains functional insulin (1)
which stimulates liver cells (1)
to convert glucose to glycogen for storage. (1)
OR
The pancreas extract contains functional insulin (1)
which stimulates body cells (1)
to take up more glucose from the blood for respiration. (1)

(d) Scientists build on the work of other scientists. (1)

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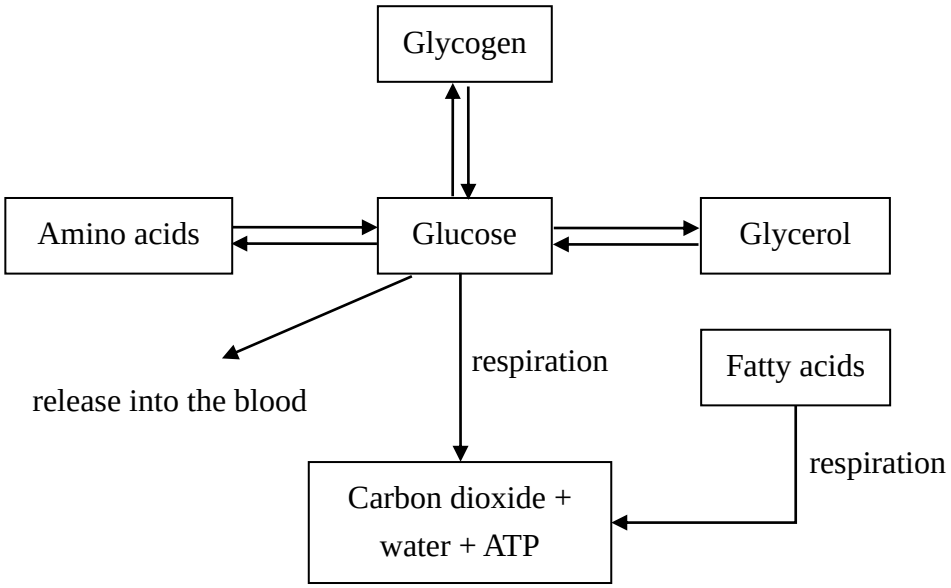
(10 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	4
(b)	Applying concepts	2
(c)	Applying concepts	3
(d)	Applying concepts	1

|!|BCELA0190000009|!| ★★

The flowchart below shows the metabolism of glucose in liver cells.



(a) The breakdown and formation of glycogen are controlled by two hormones in blood.

(i) Name the two hormones. (2 marks)

(ii) State the site of their production in the human body. (1 mark)

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(b) When excess amino acids are broken down in the liver, glucose and a nitrogen-containing compound are formed. Outline how these products are formed during amino acid breakdown. (3 marks)

(c) During prolonged fasting and the glycogen storage in the body is low, lipids in adipose tissues are hydrolysed into fatty acids and glycerol. This hydrolytic reaction is catalysed by a hormone-sensitive lipase. The fatty acids and glycerol formed are metabolized in the liver.

(i) The lipase is sensitive to the two hormones you have given in (a). Suggest the relative levels of these two hormones that can activate the lipase. (1 mark)

(ii) During fasting, glycerol can be converted to glucose which is released into the blood, while fatty acids are broken down to carbon dioxide and water in the mitochondria. State the importance of these reactions. (2 marks)

##

(a) (i) Insulin (1)
Glucagon (1)
(ii) The pancreas (1)

(b) The amino group (nitrogen-containing part) of the amino acid is converted to ammonia (1)
which is then converted to urea. (1)
The remaining part of the amino acid is converted to carbohydrates such as glucose. (1)

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(c) (i) The blood glucagon level is high while the blood insulin level is low. (1)

(ii) The conversion of glycerol to glucose helps raise the low blood glucose level to the normal range. (1)

The breakdown of fatty acids in the mitochondria can provide energy to sustain the normal functioning of cells without consuming glucose. (1)

(9 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	2
(a)(ii)	Understanding basic concepts	1
(b)	Understanding basic concepts	3
(c)(i)	Applying concepts	1
(c)(ii)	Applying concepts	2

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